

Member ID: _____

Time: _____

Rank: _____



DATABASE APPLICATIONS (240)

NATIONAL 2025

Production:

Job 1: Relationship Report	_____ (10 points)
Job 2: Maintenance Form	_____ (95 points)
Job 3: Maintenance Query	_____ (100 points)
Job 4: Vehicle Maintenance Report	_____ (70 points)
Job 5: Employee Maintenance Report	_____ (55 points)

TOTAL POINTS _____ ***(330 points)***

Industry Certification (50 minutes)
MOS Access Expert Certification

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

EXAM GUIDELINES:

1. Ensure this test booklet contains Jobs 1-5.
2. Key all jobs according to the instructions given.
3. Correct any and all formatting, spelling or grammar errors. Use the formatting guide in the *Style & Reference Manual*.
4. Your name or initials should *not* appear on any work you submit. Use your Member ID in any occasion you would normally key your reference initials.
5. In the lower **right**-hand corner of ALL work submitted (unless otherwise specified), key your Member ID and job number.
6. If you complete the event before the end of the time allotted, notify the proctor. Time may be considered a factor in determining a winner when there is a tie score.

PART ONE Instructions: Production

Once you have completed all five jobs listed below, upload all competition materials (including project files, etc.) to the National BPA Pre-submit System at presubmit.bpa.org.

DA-Job1-MemberID.pdf

DA-Job2-MemberID.pdf

DA-Job3-MemberID.pdf

DA-Job4-MemberID.pdf

DA-Job5-MemberID.pdf

PART TWO Instructions: Industry Certification

This exam requires the MOS Access Expert Certification. Please do not leave your seat or exit the room until you have completed the certification. After the certification has been administered, you are permitted to leave.

DATABASE APPLICATIONS SCORING SHEET

<i>Unless indicated otherwise, student receives all points or none.</i>	Points Possible	Points Earned
Job 1 – Relationship Report (10 points maximum)		
Report printed with relationships shown	10	
Job 2 – Maintenance Form (95 points maximum)		
Designed as instructed (10 points per field)	50	
Formatted correctly (should look like example)	15	
Field names shown with appropriate spaces (5 points each)	20	
Printed record for MaintOrder: WE2045	10	
Job 3 – Maintenance Query (100 points maximum)		
Designed as instructed (10 points per field)	60	
Sorted by Service Date (ascending)	10	
Vehicles with maintenance since July 1, 2024	10	
Proofreading/data entry (10 – 0 errors; 9 – 1 error; 7 – 2 errors; 5 – 3 errors; 0 – 4+ errors)	10	
Printed in landscape orientation, all data shows, 1 page	10	
Job 4 – Vehicle Maintenance Report (70 points maximum)		
Title is Vehicle Maintenance Report XX-XXXX-XXXX	5	
Subtitle is Professional Business Associates	5	
Correct fields are included in the report	10	
Grouped by Service Type	10	
Sorted by Service Date	10	
Only vehicles with maintenance since 7/1/24 shown	10	
Field names spaced appropriately	10	
Printed in portrait orientation, 1 page, all fields fully visible	10	
Job 5 – Employee Maintenance Report (55 points maximum)		
Title is Employee Maintenance Report XX-XXXX-XXXX	5	
Correct fields are included in the report	10	
Grouped by Employee Last Name	10	
Sorted by Service Date	10	
Appropriate spaces between field titles	10	
Printed in portrait orientation, 1 page, all fields fully visible	10	
Total Points Earned	330	

Project: Digital Solutions develops and sells software, hardware, and services. The services provided are: systems analysis, software development and design, marketing, training development, and hardware and software sales. Digital Solutions allows authorized employees to operate a DS vehicle in the course of their assigned work. Some employees may choose to operate their own vehicle while performing DS duties.

Julie Smith, Human Resources Department Manager, has asked you to create a database to keep track of all employees and their driving/vehicle records. Create a database named **National-XX-XXXX-XXXX**, and save it as directed by the administrator.

Job 1a: Employee Table

1. Create the table named **Employee XX-XXXX-XXXX** shown below. Use the same field names and assign the best data type for each field. EmployeeID is the primary key.
2. Enter all records shown below:

Employee ID	LastName	FirstName	DLNumber	VehicleNumber
ASD3021	Renick	Edna	104529483	214
CEO1456	Wells	Nancy	101142748	218
FSD3920	Rosen	Harvey	104950293	229
HRD3401	Smith	Julie	104839103	225
ITD3019	Carlson	Tom	106930583	224
MD1045	Meyer	Roger	104549288	235

Job 1b: Vehicle Table

1. Create a table named **Vehicle XX-XXXX-XXXX** shown below. Use the same field names and assign the best data type for each field. VehicleNum is the primary key (use the same data type for VehicleNum in ALL tables).
2. Enter all records shown below:

VehicleNum	VINNUM	VehicleYear	VehicleMake	VehicleModel
214	1HBH41JXMN109186	2019	GMC	Yukon
224	JHLRD77874C026456	2022	Toyota	Camry
218	1HGBH41JXMN39186	2023	Ford	Focus
235	1G1YY25R695700001	2021	Nissan	Murano
229	1G1YZ23J9P5800001	2022	Hyundai	Sonata
225	1GNCS1Z3M0115561	2023	Dodge	Challenger

Job 1c: Maintenance Table

1. Create a table named **Maintenance XX-XXXX-XXXX** shown below. Use the same field names and assign the best data type for each field. MaintOrder is the primary key.
2. Enter all records shown below:

MaintOrder	ServiceType	ServiceDate	Odometer	VehicleNum
JE2331	Brake check	6/22/2024	25603	218
TH2049	Brake check	8/28/2024	78201	229
WE2045	Brake check	10/28/2024	90241	235
HE3382	Fluid flush	2/12/2024	25603	218
IE2399	Oil change	3/26/2024	85534	214
JH2345	Oil change	8/25/2024	29874	218
UD2045	Oil change	5/21/2024	45628	224
PW2943	Oil change	7/13/2024	52156	225
CN2440	Rotate tires	5/24/2024	87234	214
EZ2883	Rotate tires	11/2/2024	49135	224
WL2055	Rotate tires	9/2/2024	58234	225
KW4301	Rotate tires	9/12/2024	53023	229

Job 1 Printout: Create a Permanent Relationship

1. Create a permanent relationship between VehicleNum in all three tables.
2. Enforce referential integrity.
3. Print the Relationship Report. **(Printout #1)**

Job 2: Maintenance Form

1. Create a form named Maintenance Form XX-XXXX-XXXX using tables/fields listed below:

Maintenance

VehicleNum

ServiceDate

Odometer

ServiceType

MaintOrder

2. Format the form as shown below:

Maintenance XX-XXXX-XXXX

Digital Solutions

VehicleNum

▼

ServiceDate

Odometer

ServiceType

MaintOrder

Thank you for keeping our roads SAFE!

We appreciate you!

3. Be sure to place colons and appropriate spaces between field titles (as shown).

Job 2: Maintenance Form (continued)

4. Using the new form, find record MaintOrder: WE2045.
5. Print the form showing only record MaintOrder: WE2045. **(Printout #2)**

Job 3: Maintenance Query

1. Create a query named **Maintenance XX-XXXX-XXXX Query** using the tables/fields listed below:

<u>Employee</u>	<u>Vehicle</u>	<u>Maintenance</u>
FirstName	VehicleMake	ServiceDate
LastName	VehicleModel	ServiceType

2. Show only the vehicles with maintenance since July 1, 2024.
3. Sort by ServiceDate (ascending).
4. Print the results in landscape orientation. All data should be visible on one page.

(Printout #3)

Job 4: Vehicle Maintenance Report

1. Create a stepped report as shown below based on the Maintenance Query created in Job 4. Your report format such as colors and placement of fields may be different, but it must:
 - a. Have the same title and subtitle.
 - b. Show the same fields.
 - c. Group by ServiceType.
 - d. Sort by LastName (ascending).
 - e. Show only vehicles with maintenance since July 1, 2024. (This information will NOT be shown on report).
 - f. Be sure to place appropriate spaces between field titles (as shown).
 - g. Have field widths/placement adjusted so all data is visible on one page in Portrait orientation.

Vehicle Maintenance Report XX-XXXX-XXXX			
Professional Business Associates			
Service Type	Employee Last Name	Vehicle Make	Vehicle Model

2. Save the report as **Vehicle Maintenance Report XX-XXXX-XXXX**.
3. Print the report in portrait orientation, on one page, with all fields visible. **(Printout #4)**

Job 5: Employee Maintenance Report

1. Create a tabular report as shown below. Your report format such as borders, colors, and placement of fields may be different, but it must:
 - a. Have the same title.
 - b. Show the same fields.
 - c. Group by Employee Last Name.
 - d. Sort by Service Date (descending).
 - e. Be sure to place appropriate spaces between field titles (as shown).
 - f. Have field widths/placement adjusted so all data is visible on one page in Landscape orientation.

Employee Maintenance Report XX-XXXX-XXXX		
Employee	Service Date	Service Type

2. Save the report as **Employee Maintenance Report XX-XXXX-XXXX**.
3. Print the report in portrait orientation, on one page, with all fields visible. **(Printout #5)**

When finished, place your printouts in the following order:

1. Relationship Report
2. Vehicle Form
3. Maintenance Query
4. Vehicle Maintenance Report
5. Employee Maintenance Report
6. All incomplete/incorrect printed pages (draw a line through each one).